

PREET PATEL

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PROFESSIONAL EXPERIENCE

U.S. Patent & Trademark Office

Oct 2024 – Present

Patent Examiner (Physics)

- **Technical Due Diligence:** Evaluate the novelty and non-obviousness of complex intellectual property in quantum computing, photonic circuits, and fiber optics.
- **Analytical Assessment:** Synthesize vast amounts of technical literature to reach legal determinations — a skill directly transferable to private equity due diligence and market landscape analysis.

Global Logic Inc. (Google RMI Team)

Jul 2024 – Oct 2024

Data Quality Analyst (“Superrater”)

- **AI Model Curation:** Acted as a key AI Evaluator for Google’s Gemini and Magi models, performing high-level evaluations of LLM outputs to ensure alignment with quality and safety guidelines.
- **Strategic Prompt Engineering:** Developed and refined complex prompts to test model boundaries, directly influencing the iterative improvement of model accuracy and reasoning.
- **Data Integrity:** Identified edge cases and data anomalies to maintain high-fidelity training sets for large-scale machine intelligence deployments.

University of California, Davis

Mar 2021 – Sept 2023

Computational Astrophysicist

- **Big Data Pipeline Engineering:** Managed and parsed petabytes of HDF5 simulation data across national supercomputing clusters, optimizing analysis pipeline runtime by 50%.
- **Predictive Modeling:** Applied supervised learning and time-series analysis to model the evolution of complex systems, resulting in a first-author publication in MNRAS.
- **HPC Optimization:** Implemented parallel processing (OpenMP/MPI) and CUDA-based optimizations to simulate n-body dynamics with 10,000+ data points per simulation.

TECHNICAL PROJECTS

Quantitative Logic & Element Tracers

Python / HPC • GitHub

- Streamlined modular data pipelines into a single class structure, achieving a 50%+ increase in computational speed.
- Packaged the solution for deployment on Linux-based clusters, providing a scalable framework for 200+ collaborators to perform sensitivity analysis on dynamic datasets.

Quantitative Analysis: MCMC & Maximum Likelihood

Python • GitHub

- Built a data analysis pipeline to analyze raw observational datasets using SQL queries and Markov Chain Monte Carlo (MCMC) simulations.
- Engineered a log-likelihood minimization framework to isolate signals within noisy datasets, identifying key sources for system calibration.

EDUCATION

M.S. Physics — University of California, Davis

2020 – 2023

B.S. Physics & B.S. Astrophysics, Minor in Statistics — University of Michigan

2015 – 2019

TECHNICAL SKILLS

Languages & Tools: Python (Pandas, Polars, SciPy, Matplotlib, Flask), SQL, R, C++, Git, Bash, Linux/Unix.

Data Science: Machine Learning (PyTorch, TensorFlow, Scikit-learn), Time-Series Analysis (ARIMA, LSTM), Regression, Feature Engineering, EDA.

Operations: High-Performance Computing (CUDA, MPI/OpenMP), Excel, Graphic Design (Photoshop).